# Self-Supervised Image Segmentation from a Single Image

## Problem Definition

This project addresses the problem of image segmentation without any ground truth labels, by using a self-supervised learning approach. The input is a single RGB image, and the output is a segmentation map where each pixel is assigned a region label. This task is important because obtaining labelled segmentation data is generally labour intensive and expensive. Furthermore, self-supervised methods learn from the image itself, making them suitable for scenarios where manual annotations are not available. A successful outcome involves generating a visually semantically meaningful segmentation map, which can be evaluated qualitatively by observing the continuity of object regions and by examining cluster separability.

$$\text{Input image: } I \in R^{H \times W \times 3}$$

$$\text{Segmentation map: } S \in R^{H \times W}$$

## Method

The code segments an image into meaningful regions by learning how different parts of the image relate to each other, using contrastive self-supervised learning. It starts by sampling random patches of size 32×32 pixels from the input image. Each patch is then augmented twice using techniques like cropping, color jittering, and flipping to create a pair of similar-looking patches. These pairs act as training examples to help the model understand what makes two image regions similar.

To process these patches, a small convolutional neural network (CNN) is used to turn each one into a lower-dimensional feature vector or embedding. The CNN is kept fairly simple, consisting of two convolutional layers with Rectified Linear Unit (ReLU) activations, followed by an adaptive average pooling layer. This setup helps the model extract key features from the patches in a compact form while keeping the architecture lightweight and efficient.

Once the patches are encoded, a contrastive learning technique is applied using the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss. This loss function encourages the model to pull the embeddings of augmented versions of the same patch closer together, while pushing apart those of different patches. This helps the model learn strong and consistent representations of visual features without needing any manual labels.

$$\mathrm{l(i,j)} = -\log\left(\frac{\exp\left(\mathrm{sim}\left(\mathrm{z_i}, \mathrm{z_j}\right)/\tau\right)}{\sum_{\mathrm{k}=1}^{2\mathrm{N}} 1_{\mathrm{k}\neq\mathrm{i}} \exp(\mathrm{sim}(\mathrm{z_i}, \mathrm{z_k})/\tau)}\right)$$

The implementation of the contrastive loss function used in this project was guided by the official ***Simclr google-research repository*** [1] which provided a clear reference for how to compute the Normalized Temperature-scaled Cross Entropy (NT-Xent) loss in practice. Studying this implementation helped in understanding how to structure the similarity matrix, apply masking for positive and negative pairs, and scale the logits appropriately for stable training.

After training the encoder, the full image is broken into overlapping patches using a sliding window. These patches are passed through the trained encoder to generate their embeddings. K-Means clustering is then used to group these embeddings into clusters, each representing a different region in the image. The final segmentation map is built by averaging the cluster labels in the overlapping areas, producing a clean and coherent segmentation of the image.

## Results

The experiment was conducted on a single image of building John Curtin School of Medical Research at ANU provided by the ANU Faculty, with 500 random patches used for training. The encoder was trained for 10 epochs using the Adam optimizer and a learning rate of 1e-3.[2] After training, 32x32 patches were extracted every 16 pixels to create overlapping embeddings for segmentation. The CNN encoder used in this work is deliberately simple, comprising two convolutional layers and pooling, chosen to reduce overfitting and ensure computational efficiency. However, this simplicity may limit the capacity to capture higher-level semantics or texture patterns, especially in more complex scenes. Similarly, the choice of 32×32 patch size represents a trade-off, large enough to contain contextual cues, but small enough to provide detailed spatial information.

As shown in ***Figure 1*** training over 10 epochs revealed a steady decline in the NT-Xent loss, indicating that the encoder progressively learned more discriminative representations. However, the rate of improvement diminished toward the later epochs, suggesting a point of diminishing returns. This highlights the trade-off between compute time and representation quality, especially in single-image training scenarios. Training over 50 epochs was also considered but the rate of improvement diminished at 6.1 as shown in ***Figure 4***. Although no ground truth is available for numerical evaluation, clustering quality was indirectly assessed through visual coherence and cluster compactness. For example, using 10 clusters resulted in visually distinct regions corresponding to different parts of the image (e.g., sky, windows, building walls). As shown in ***Figure 2***, the segmented image displays clear separation between structural components of the scene. Regions such as the roof, sky, and glass panes form coherent clusters, indicating that the encoder captured semantically meaningful features despite having no labels.

Furthermore, to explore the effect of cluster granularity on segmentation quality, multiple test cases were evaluated using different numbers of clusters being 3, 5, 7, and 10 as shown in ***Figure 3***. As the number of clusters increased, the segmentation maps revealed progressively finer details. While fewer clusters (e.g., 3 or 5) grouped large regions effectively, they failed to distinguish smaller visual components. In contrast, using 10 clusters produced the most visually coherent and semantically rich segmentation, successfully isolating features like windows, roofs, and sky regions. This suggests that higher cluster counts can better capture structural variation in the image, though overly high values may risk over segmentation.

## Reflection

The work done highlights the potential of self-supervised methods for segmentation tasks in low-data regimes. It could benefit areas like medical imaging or satellite imagery, where manual labelling is costly or infeasible. However, the approach may be less effective on cluttered or highly textured scenes, and there is a risk of amplifying data biases if the encoder overfits to low-level patterns.

## Conclusion

Overall, the project demonstrated that self-supervised contrastive learning can effectively segment a single image into semantically meaningful regions. Future improvements could include multi-scale training, spatial consistency constraints, or extension to video inputs for temporal coherence.

## Appendix

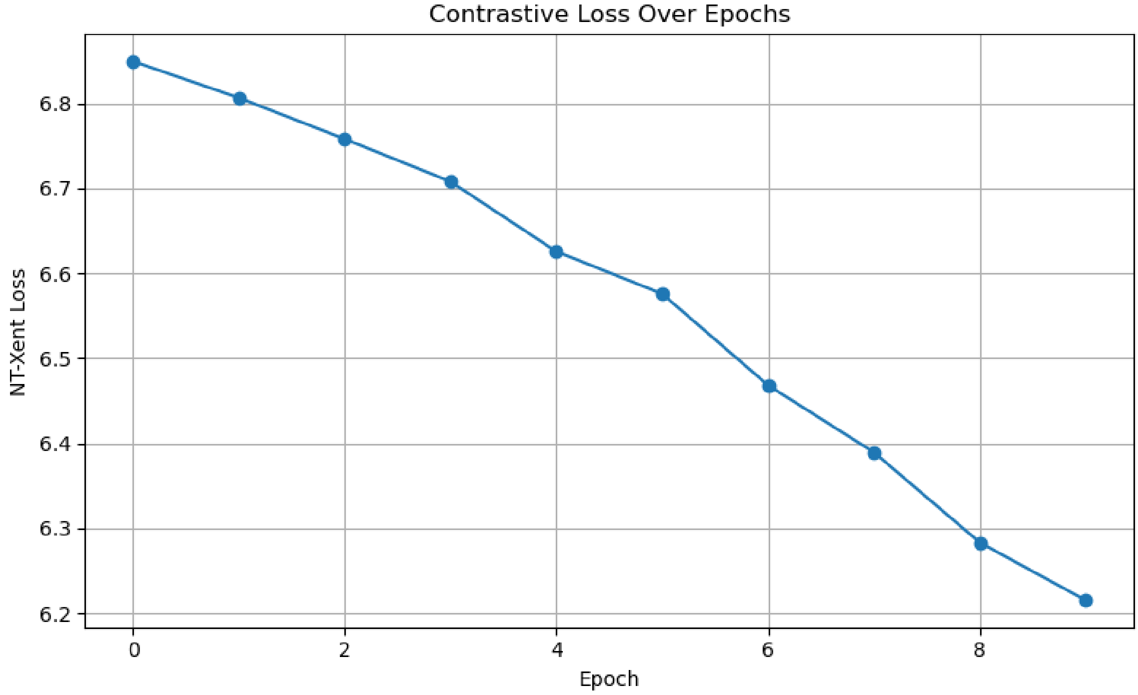


***Figure 1:*** *This plot illustrates the NT-Xent loss across training epochs for the contrastive self-supervised learning model.*

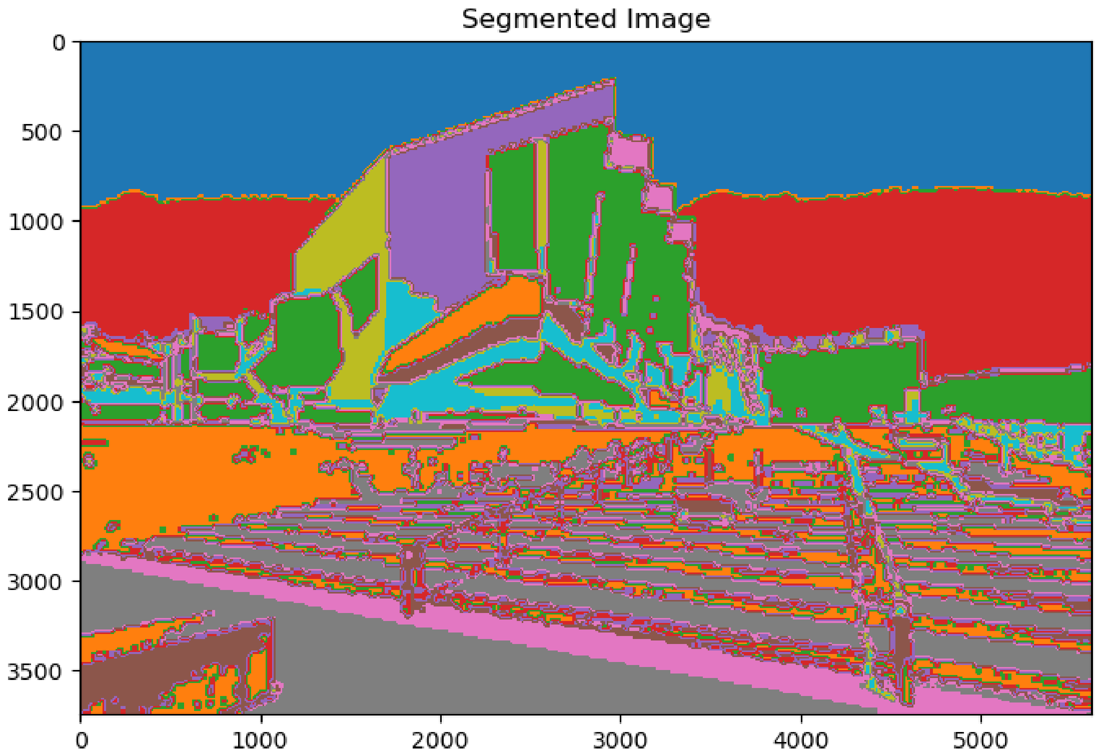


***Figure 2:*** *Visualisation of the final segmentation map using 10 clusters. Colour regions correspond to distinct segments identified by the model. Note the consistent grouping of similar textures and semantic parts.*

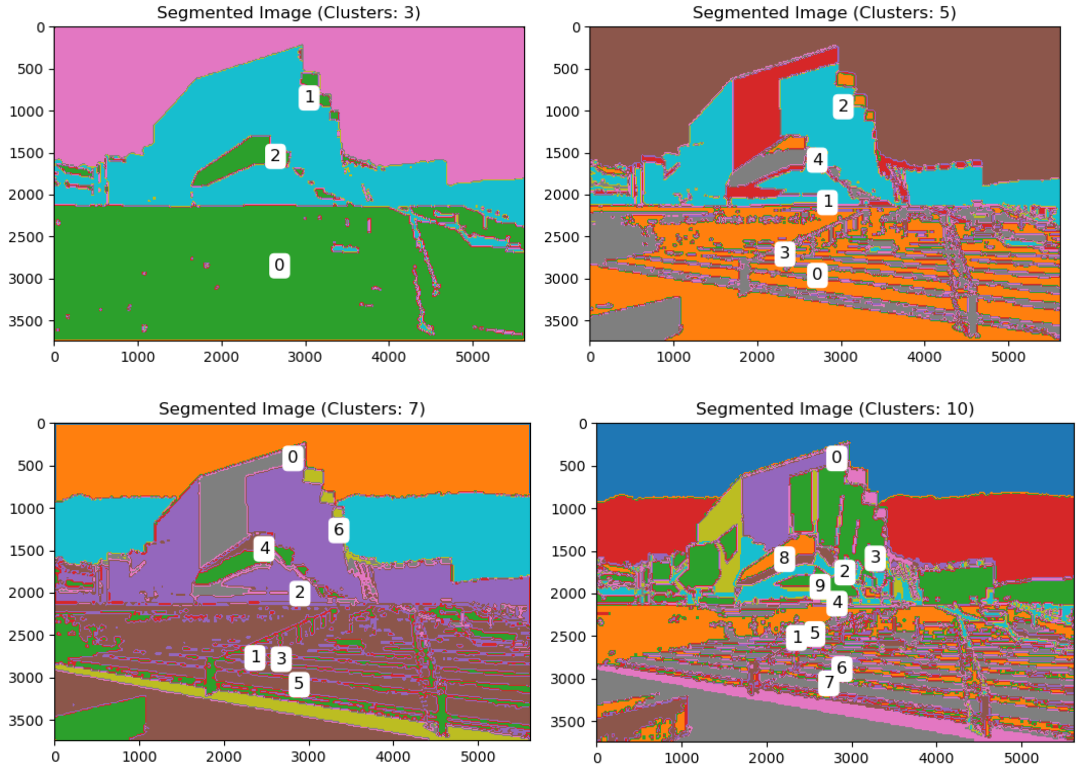


***Figure 3:*** *Visualisation of the final segmentation map using 3,5,7,10 clusters. Colour regions correspond to distinct segments identified by the model. Note the consistent grouping of similar textures and semantic parts.*

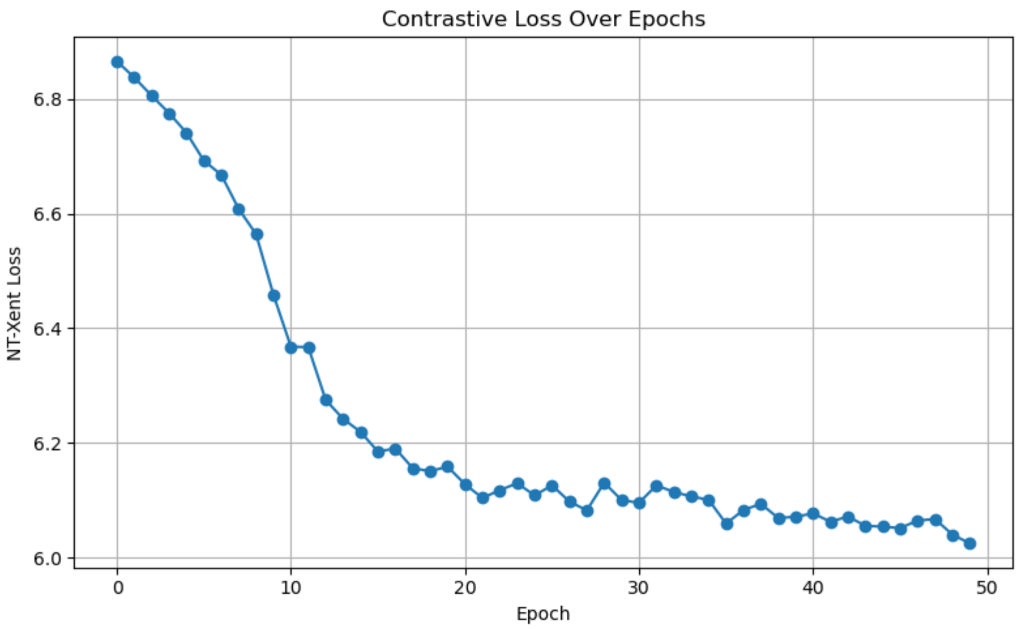


***Figure 4:*** *This plot illustrates the NT-Xent loss across training 50 epochs for the contrastive self-supervised learning model.*

**References**

1. Chen, T., Kornblith, S., Norouzi, M. and Hinton, G., 2020. *A Simple Framework for Contrastive Learning of Visual Representations*. [online] GitHub. Available at: https://github.com/google-research/simclr/tree/bfe07eed7f101ab51f3360100a28690e1bfbf6ec [Accessed 12 Apr. 2025].

2. Brownlee, J., 2020. *Understand the Dynamics of Learning Rate on Deep Learning Neural Networks*. [online] Machine Learning Mastery. Available at: https://machinelearningmastery.com/understand-the-dynamics-of-learning-rate-on-deep-learning-neural-networks/ [Accessed 14 Apr. 2025].

3. Zhang, J., Zhang, W., Li, X., Cao, Y., Zhou, J. and Luo, P., 2023. *SEED: Self-supervised Distillation for Visual Representation*. arXiv preprint arXiv:2309.10511. Available at: https://arxiv.org/pdf/2309.10511 [Accessed 10 Apr. 2025].